 

# ANZALI, Iran, Islamic Republic of

WMO: **407180**

Lat: **37.4797N** Lon: **49.4579E** Elev: **-24** StdP: **101.61** Time Zone: **3.50 (IRN)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **2** | **1.3** | **2.8** | **-1.5** | **3.3** | **5.4** | **0.1** | **3.8** | **4.9** | **15.3** | **3.6** | **14.0** | **4.3** | **4.3** | **330** | **0.376** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **8** | **5.0** | **31.6** | **25.6** | **30.7** | **25.3** | **29.9** | **25.0** | **27.0** | **30.1** | **26.4** | **29.6** | **25.8** | **29.1** | **3.8** | **60** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **26.0** | **21.3** | **29.4** | **25.4** | **20.5** | **29.0** | **24.7** | **19.7** | **28.5** | **84.8** | **30.1** | **82.2** | **29.7** | **79.7** | **29.2** | **30.5** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **12.2** | **9.9** | **7.9** | DB | **-0.4** | **33.0** | **2.3** | **1.2** | **-2.0** | **33.9** | **-3.3** | **34.7** | **-4.6** | **35.4** | **-6.2** | **36.3** | (4) |
| (5) | | | | WB | **-0.6** | **27.8** | **2.0** | **0.9** | **-2.1** | **28.5** | **-3.2** | **29.1** | **-4.4** | **29.6** | **-5.8** | **30.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **16.9** | **8.1** | **7.7** | **9.9** | **13.7** | **19.6** | **24.2** | **26.5** | **26.6** | **23.3** | **18.8** | **13.8** | **10.2** | (6) |
| (7) | | DBStd | | **7.36** | **2.80** | **2.83** | **2.75** | **2.86** | **2.69** | **2.37** | **1.91** | **1.95** | **2.39** | **2.68** | **3.00** | **2.90** | (7) |
| (8) | | HDD10.0 | | **214** | **69** | **73** | **32** | **3** | **0** | **0** | **0** | **0** | **0** | **0** | **5** | **32** | (8) |
| (9) | | HDD18.3 | | **1461** | **318** | **298** | **264** | **142** | **18** | **1** | **0** | **0** | **1** | **27** | **140** | **254** | (9) |
| (10) | | CDD10.0 | | **2738** | **11** | **8** | **28** | **114** | **298** | **427** | **512** | **514** | **398** | **273** | **118** | **36** | (10) |
| (11) | | CDD18.3 | | **943** | **1** | **0** | **2** | **3** | **58** | **177** | **253** | **256** | **149** | **41** | **3** | **1** | (11) |
| (12) | | CDH23.3 | | **7636** | **3** | **2** | **18** | **19** | **118** | **1218** | **2581** | **2693** | **903** | **68** | **11** | **2** | (12) |
| (13) | | CDH26.7 | | **1973** | **0** | **0** | **6** | **5** | **6** | **234** | **738** | **857** | **120** | **5** | **3** | **0** | (13) |
| (14) | Wind | WSAvg | | **2.8** | **3.2** | **3.0** | **2.9** | **2.6** | **2.6** | **2.7** | **2.5** | **2.6** | **2.8** | **2.9** | **2.9** | **2.9** | (14) |
| (15) | Precipitation | PrecAvg | | **1308** | **110** | **97** | **91** | **73** | **44** | **44** | **58** | **78** | **174** | **204** | **193** | **141** | (15) |
| (16) | | PrecMax | | **2142** | **277** | **229** | **141** | **173** | **102** | **162** | **171** | **407** | **451** | **404** | **446** | **307** | (16) |
| (17) | | PrecMin | | **898** | **32** | **13** | **4** | **8** | **15** | **0** | **5** | **0** | **17** | **0** | **0** | **0** | (17) |
| (18) | | PrecStd | | **241** | **64** | **54** | **37** | **45** | **23** | **39** | **44** | **77** | **98** | **106** | **103** | **82** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **21.2** | **21.5** | **25.8** | **26.4** | **27.4** | **32.0** | **32.9** | **32.5** | **30.2** | **26.8** | **24.5** | **22.9** | (19) |
| (20) | | | MCWB | | **11.4** | **11.7** | **14.3** | **16.6** | **21.4** | **24.7** | **25.8** | **26.0** | **25.5** | **21.8** | **15.6** | **13.1** | (20) |
| (21) | | 2% | DB | | **15.4** | **14.6** | **17.2** | **20.4** | **25.8** | **30.0** | **31.6** | **31.6** | **29.0** | **25.0** | **20.5** | **18.1** | (21) |
| (22) | | | MCWB | | **10.0** | **10.2** | **13.1** | **16.2** | **21.6** | **24.0** | **25.7** | **25.9** | **24.8** | **22.0** | **17.3** | **12.3** | (22) |
| (23) | | 5% | DB | | **12.3** | **11.8** | **14.3** | **18.5** | **24.6** | **28.9** | **30.6** | **30.8** | **28.0** | **23.9** | **19.0** | **15.2** | (23) |
| (24) | | | MCWB | | **10.0** | **9.8** | **12.0** | **15.8** | **21.0** | **23.6** | **25.2** | **25.7** | **24.2** | **21.2** | **16.8** | **13.0** | (24) |
| (25) | | 10% | DB | | **11.1** | **10.7** | **12.7** | **17.3** | **23.5** | **27.9** | **29.8** | **30.1** | **27.1** | **22.9** | **17.9** | **13.7** | (25) |
| (26) | | | MCWB | | **9.6** | **9.3** | **11.1** | **15.2** | **20.5** | **23.2** | **24.8** | **25.5** | **23.7** | **20.4** | **16.1** | **12.1** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **12.6** | **12.5** | **15.1** | **18.4** | **23.2** | **26.1** | **28.2** | **27.7** | **26.5** | **23.6** | **19.3** | **15.2** | (27) |
| (28) | | | MCDB | | **17.9** | **18.9** | **23.4** | **22.3** | **25.5** | **29.8** | **31.1** | **30.8** | **29.2** | **25.5** | **20.8** | **17.2** | (28) |
| (29) | | 2% | WB | | **11.1** | **10.7** | **13.2** | **17.0** | **22.3** | **25.2** | **27.0** | **27.1** | **25.5** | **22.5** | **18.0** | **13.9** | (29) |
| (30) | | | MCDB | | **13.5** | **12.8** | **16.2** | **19.2** | **24.7** | **28.7** | **30.4** | **30.1** | **28.2** | **24.4** | **19.6** | **15.7** | (30) |
| (31) | | 5% | WB | | **10.3** | **10.1** | **12.1** | **16.1** | **21.5** | **24.6** | **26.2** | **26.6** | **24.7** | **21.6** | **17.0** | **12.9** | (31) |
| (32) | | | MCDB | | **11.9** | **11.5** | **14.2** | **18.2** | **23.9** | **27.9** | **29.6** | **29.7** | **27.4** | **23.3** | **18.6** | **15.0** | (32) |
| (33) | | 10% | WB | | **9.8** | **9.5** | **11.2** | **15.3** | **20.8** | **23.9** | **25.5** | **26.1** | **24.0** | **20.7** | **16.1** | **12.1** | (33) |
| (34) | | | MCDB | | **11.0** | **10.7** | **12.8** | **17.2** | **23.0** | **27.0** | **28.9** | **29.3** | **26.7** | **22.5** | **17.7** | **13.7** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **4.0** | **3.7** | **3.7** | **3.8** | **4.0** | **4.6** | **4.8** | **5.0** | **4.8** | **4.7** | **4.5** | **4.1** | (35) |
| (36) | | 5% DB | MCDBR | | **7.1** | **6.8** | **8.0** | **6.6** | **5.4** | **6.1** | **6.0** | **6.1** | **5.8** | **5.8** | **6.3** | **6.4** | (36) |
| (37) | | | MCWBR | | **4.0** | **4.0** | **4.1** | **3.3** | **2.9** | **3.2** | **3.3** | **3.7** | **3.5** | **3.7** | **4.1** | **4.0** | (37) |
| (38) | | 5% WB | MCDBR | | **5.1** | **5.4** | **7.0** | **5.5** | **4.7** | **5.2** | **5.2** | **5.1** | **5.0** | **5.1** | **5.3** | **5.3** | (38) |
| (39) | | | MCWBR | | **3.4** | **3.4** | **3.9** | **3.2** | **3.0** | **3.2** | **3.4** | **3.4** | **3.5** | **3.7** | **4.0** | **3.7** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.396** | **0.440** | **0.492** | **0.577** | **0.560** | **0.559** | **0.602** | **0.576** | **0.552** | **0.505** | **0.434** | **0.398** | (40) |
| (41) | | taud | | | **2.187** | **2.070** | **1.971** | **1.822** | **1.907** | **1.937** | **1.860** | **1.917** | **1.952** | **2.026** | **2.149** | **2.201** | (41) |
| (42) | | Ebn at Noon | | | **758** | **769** | **767** | **725** | **748** | **748** | **710** | **718** | **706** | **694** | **707** | **724** | (42) |
| (43) | | Edh at Noon | | | **110** | **139** | **169** | **207** | **194** | **188** | **201** | **186** | **171** | **144** | **114** | **102** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **1.91** | **2.44** | **3.32** | **4.32** | **5.68** | **6.34** | **6.16** | **5.50** | **4.13** | **2.87** | **2.05** | **1.67** | (44) |
| (45) | | RadStd | | | **0.21** | **0.35** | **0.39** | **0.53** | **0.39** | **0.39** | **0.47** | **0.62** | **0.34** | **0.39** | **0.26** | **0.20** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **+0.33** | **N/A** | **N/A** | **+1.01** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (8 neighbors) | **+0.40** | **N/A** | **N/A** | **+0.98** | **+0.43** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page